INCH-POUND

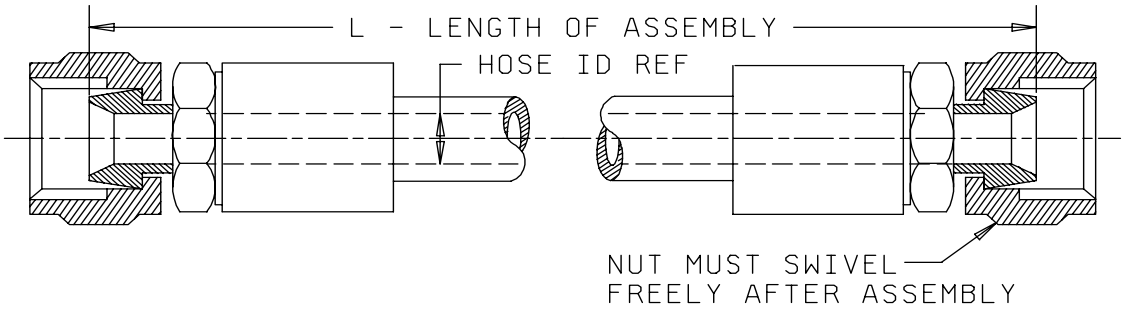
MS500083E
17 May 2006
SUPERSEDING
MS500083D
16 August 2002

DETAIL SPECIFICATION SHEET

HOSE ASSEMBLY, NONMETALLIC: HYDRAULIC AND PNEUMATIC, HIGH PRESSURE, FEMALE 37° FLARED TUBE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-13531.



NOTES:
1. All undimensioned pictorial representations are for reference purposes only.
2. Metric equivalents are given for information only.
3. Dimension “L” is the defined length of the hose assembly.
4. Hose assemblies shall be specified in the following length increments:
   Under 24 inches (0.61 m) ½ inch (12.70 mm)increments.
   24 to 36 inches (0.61 to 0.91 m) - 1-inch (25.40 mm) increments.
   Over 36 inches (0.91 m)- in not less than 2-inch (50.80 mm) increments.
5. Tolerances:
   ±1/8 inch (.125 inch) (38.10 mm) for length under 24 inches (0.61 m).
   ±1/4 inch (.250 inch) (76.20 mm) for the length from 24 to 36 inches (0.61 to 0.91 m).
   ±1% for length over 36 inches (0.91 m).

FIGURE 1. General configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T20:47Z
Check the source to verify that this is the current version before use.

TABLE I. Hose and fitting thread sizes.

Dash number		Hose ID reference		Hose OD reference				Fitting thread reference
				Class A		Class B		
Permanent fittings	Reuseable fittings	Inches	mm	Inches	mm	Inches	mm	
3	33	.187	5.0	.625	15.9	.535	13.6	3/8-24
4	34	.250	6.5	.687	17.5	.593	15.0	7/16-20
6	35	.375	9.5	.843	21.4	.754	19.0	9/16-18
8	36	.500	12.5	.968	24.6	.874	22.2	3/4-16
10	37	.635	15.9	1.093	27.8	1.008	25.6	7/8-14
12	38	.750	19.0	1.250	31.8	1.156	29.4	1-1/16-12
16	39	1.000	25.5	1.562	39.7	1.500	38.1	1-5/16-12
20	40	1.250	32.0	2.000	51.0	1.867	47.4	1-5/8-12
24	41	1.500	38.0	2.250	57.1	2.148	54.5	1-7/8-12
32	42	2.000	51.0	2.750	70.0	2.648	67.3	2-1/2-12

REQUIREMENTS

Design and configuration: See figure 1 and table I.

Hose material: Type II or III, class A or B.

Hose and fittings shall be in accordance with MIL-DTL-13531.

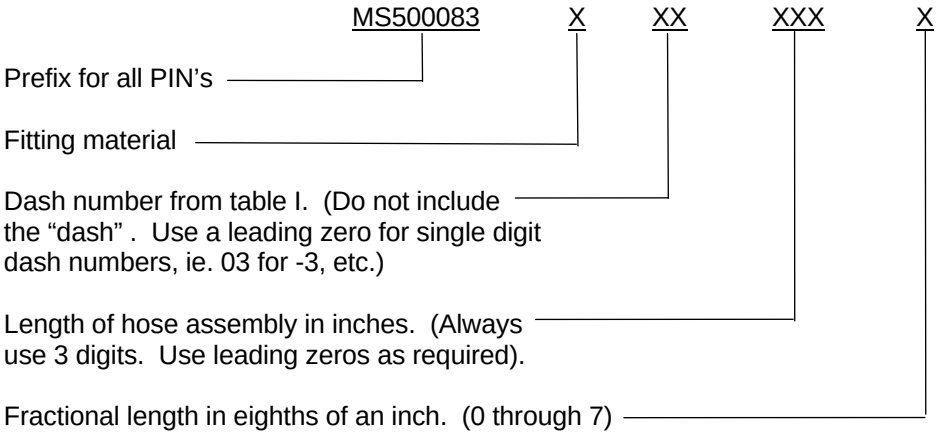
Fittings: Dimensions in accordance with SAE-J516 female 37°, flared type, swivel.

Fitting material and finish shall be in accordance with MIL-DTL-13531 and designated as follows:

A - Steel - cadmium plated
S - Stainless steel.
Z - Steel - zinc plated. (Zinc plating is not for use in aircraft).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T20:47Z
Check the source to verify that this is the current version before use.

Part or Identifying Number (PIN). The PIN is to be used for hose assemblies acquired to this specification. Example is shown below:



Changes from previous issue: The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-DTL-13531, this document references the following:

SAE-J516

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-2005-010)

Review activities:
Army - AR, AV, CR4, MI
Navy - AS, MC, SA,
Air Force - 11, 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T20:47Z
Check the source to verify that this is the current version before use.